

"Best Approx in Hilbert Space"

Idea: Building up functions w/ simple pieces:

$$\textcircled{*} \quad \begin{matrix} [] & & [] & & [] & & [] \\ f(x) & \approx & c_1 e_1(x) & + & c_2 e_2(x) & + \dots & + c_n e_n(x) \end{matrix}$$

Q1: How to ^(compute) choose $c_1, \dots, c_n$? $\begin{bmatrix} f \\ 1 \end{bmatrix} = [A] \begin{bmatrix} c \\ 1 \end{bmatrix}$

Q2: How to analyze approximation error?

Idea: Develop "linear algebra for functions"

$\Rightarrow$ Hilbert spaces for functions

$\Rightarrow$ "Inner product space" + "Completeness"

$$f + \alpha g$$

- Linear combos of functions

$$\|f\|$$

- Measure "size" of a function (norm)

$$\|f - g\|$$

- Measure "distance" between functions

$$\langle f, g \rangle$$

- Measure "angles" between functions

$$\|f\| = \sqrt{\langle f, f \rangle}$$

(inner product)

- $f = c_1 e_1 + c_2 e_2 + c_3 e_3 + \dots \in H$

as long as $\|f\| < \infty$,

Best Approx: Choose $c_1, \dots, c_n$ so that

$\|f - (c_1 e_1 + c_2 e_2 + \dots + c_n e_n)\|$ is minimized.

Here, $\{e_1, e_2, \dots, e_n\}$ are an orthonormal set.

$$\langle e_i, e_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

Throughout, $f \in H = \text{Hilbert space}$, $V = \text{span}\{e_1, \dots, e_n\} \subseteq H$

Example: $H = \mathbb{R}^m$ and $V \subseteq \mathbb{R}^m$, with $n < m$.

$$\begin{aligned} \|f - (c_1 e_1 + c_2 e_2 + \dots + c_n e_n)\|_2 &= \left\| \begin{bmatrix} 1 \\ f \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & & 1 \\ e_1 & \dots & e_n \\ 1 & & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right\|_2 \\ &= \| \xi - E c \|_2 \end{aligned}$$

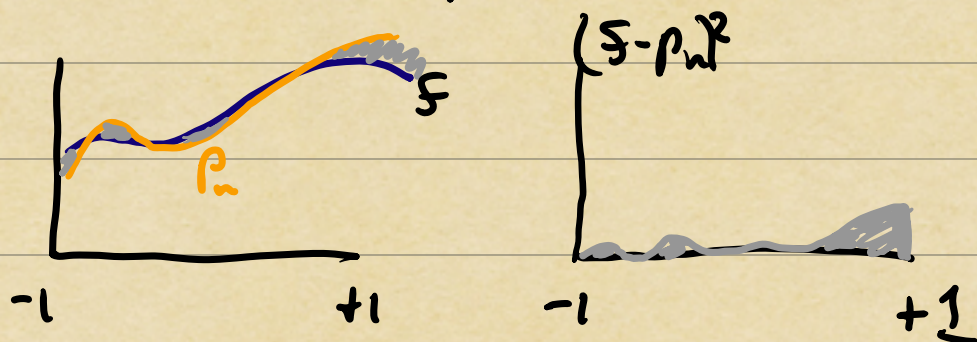
Q: How to solve finite-dimensional least-squares,
 $\Rightarrow$ normal equations ($E^T E c = E^T \xi$)
 $\Rightarrow$ orthogonal projection ($E c = E E^T \xi$)

$$\begin{bmatrix} \equiv \\ \vdots \\ \vdots \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

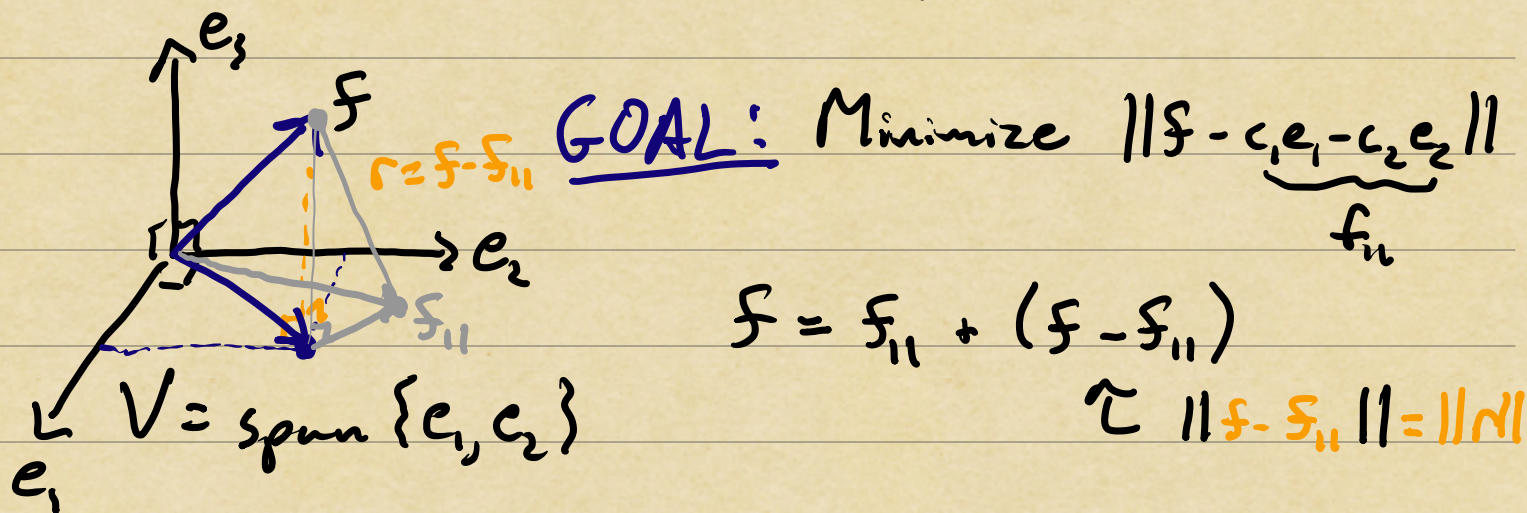
If $\langle e_i, e_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$, then $E^T E = I$

Example: $H = L^2(-1,1)$ and $V = P_n = \text{span}\{1, x, \dots, x^n\}$
 $L^2 = \{f: [-1,1] \rightarrow \mathbb{R} \mid \int_{-1}^1 |f(x)|^2 dx < \infty\}$

minimize $\int_{-1}^1 |f(x) - (c_0 + c_1 x + c_2 x^2 + \dots + c_n x^n)|^2 dx$



The "Picture" behind best Approx.



$$r = r_{||} + r_{\perp} \quad \text{where} \quad \langle r_{||}, r_{\perp} \rangle = 0$$

$$\|r\|^2 = \|r_{||}\|^2 + \|r_{\perp}\|^2 \Rightarrow$$

choose c_1, c_2 so that $r_{||} = 0$

Idea: Write $r = f - f_{||} = \tilde{r}_1 e_1 + \tilde{r}_2 e_2 + \tilde{r}_3 e_3 + \dots$

$$f_{11} = c_1 e_1 + c_2 e_2, \quad f = \hat{f}_1 e_1 + \hat{f}_2 e_2 + \hat{f}_3 e_3 + \dots$$

$$r = f - f_{11} = \underbrace{(\hat{f}_1 - c_1) e_1 + (\hat{f}_2 - c_2) e_2}_{r_{11}} + \underbrace{\hat{f}_3 e_3 + \dots}_{r_{\perp}}$$

So, choose $c_1 = \hat{f}_1$ and $c_2 = \hat{f}_2$

Orthonormal Basis

An orthonormal basis for a subspace $V_N \subset H$, with dimension $N < \infty$, is a basis $\{q_1, \dots, q_N\} \subset V_N$ such that $\langle q_i, q_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$.

Define: The orthogonal projection of $f \in H$ onto V_N is

$$P_N f = \sum_{k=1}^N \langle f, q_k \rangle q_k.$$

Claim: It satisfies

- i) $f \mapsto P_N f$ is linear ($P_N(f + \alpha g) = P_N f + \alpha P_N g$)
- ii) $P_N f = f$ when $f \in V_N$
- iii) $P_N f = 0$ when $f \perp V_N$
- iv) $\|P_N f\| \leq \|f\|$ for $f \in H$.